OBJECTS OF D(R) WITH COHOMOLOGY ONLY IN DEGREES MINUS ONE AND ZERO HAVE TWO-TERM FREE REPRESENTATIONS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof span is a two-part proof, each part opened by 'Proof of (1).' and 'Proof of (2).'; both parts are delivered verbatim, together with the author's \medskip\noindent separator. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Existence of left resolutions of complexes by complexes of free modules, vanishing in high degrees. invoked at proof lines 22319-22321. Produces the free complex whose truncation represents K.
- Pseudo-coherence and the existence of finite free resolutions of finite complexes over a Noetherian ring. invoked at proof lines 22325-22327. Produces the finite free model used in part (2).
- Truncation of complexes in the derived category. used at proof lines 22318 and 22322 and 22328. Confines the representing complex to the two degrees admitted by the cohomology hypothesis.

Source reference. Original lemma 05T7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7471–7471. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 16048: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16048–16048. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let K be an object of $D(R)$ with $H^i(K) = 0$ for $i \notin \{-1, 0\}$. Then*

(1) *K can be represented by a two term complex $K^{-1} \to K^0$ with K^0 a free module, and*

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(2) *if R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$, then K can be represented by a two term complex $K^{-1} \to K^0$ with K^0 a finite free module and K^{-1} finite.*

Proof. Proof of (1). Suppose K is given by the complex of modules $M^\bullet$. We may first replace $M^\bullet$ by $\tau_{\leq 0}M^\bullet$. Thus we may assume $M^i = 0$ for $i > 0$, Next, we may choose a free resolution $P^\bullet \to M^\bullet$ with $P^i = 0$ for $i > 0$, see Derived Categories, Lemma 05T7. Finally, we can set $K^\bullet = \tau_{\geq -1}P^\bullet$.

Proof of (2). Assume R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$. By Lemma 16048 we can choose a quasi-isomorphism $F^\bullet \to M^\bullet$ with $F^i = 0$ for $i > 0$ and F^i finite free. Then we can set $K^\bullet = \tau_{\geq -1}F^\bullet$. □